COMPLEX ENGINEERING PROBLEM

TITLE

Secure Pull Request Workflow

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DEPARTMENT

COMPUTER ENGINEERING

ACADEMIC YEAR

2026–2027

SEMESTER

5TH

SUBJECT

OPEN SOURCE TECHNOLOGIES(01CE0526)

CLASS

5EC6

BATCH

A

COMPLEX ENGINEERING PROBLEM - REPORT

RUBRICS FOR COMPLEX ENGINEERNG PROBLEM

Sr. No.	Assessment Criteria	Excellent	Good	Satisfactory	Poor	Marks
1	Problem Analysis & Open Source Technology Selection	3 Marks: Clearly identifies the engineering problem, analyzes requirements and constraints, and selects appropriate Open Source technologies with proper justification.	2 Marks: Good problem analysis with suitable technology selection and minor gaps.	1 Mark: Basic analysis with limited justification of technology selection.	0 Marks: Inadequate problem analysis and inappropriate technology selection.	3
2	System Design, Implementation Strategy & Technical Analysis	4 Marks: Well-defined system architecture, appropriate technology stack, database, deployment strategy, version control, and comprehensive technical analysis including scalability, reliability, and security.	3 Marks: Good system design and technical discussion with minor omissions.	1–2 Marks: Basic architecture and limited technical analysis.	0 Marks: Poor or missing system design and technical analysis.	4
3	Testing, Documentation & References	3Marks: Comprehensive testing strategy, well-organized report, IEEE references, appropriate diagrams, and clear conclusions.	2 Marks: Good documentation with minor formatting or testing gaps.	1 Mark: Basic report with limited testing or references.	0 Marks: Incomplete report with poor documentation and missing references.	3
	Total					10

Sr. No.	Complex Engineering Problem Activity / Problem Title	COs	Date	R1(4)	R2(3)	R3(4)	Marks (10)	Signature
1	Secure Pull Request Workflow	CO2						

Notes / Instructions

1. CEP activity must be completed in a group of maximum 3 students.
2. Students must strictly follow the given deadlines for submission.
3. The report must be prepared in the prescribed format and submitted as a DOCX file on Google Classroom.
4. A hard copy of the report must be submitted as discussed.
5. The report should include problem analysis, design/architecture, implementation, results, and challenges faced.
6. All group members must contribute equally and be prepared for evaluation/presentation.
7. Plagiarism or copying from other groups will not be accepted.
8. Each group member must understand the complete solution and should be able to explain their contribution during evaluation.

Ms. Priyanka Mangi
Subject Faculty

COMPLEX ENGINEERING PROBLEM - REPORT

Secure Pull Request Workflow
Course Name: OPEN SOURCE TECHNOLOGIES(01CE0526)
Co's Mapped - CO2
SDG's Mapped - SDG 16, SDG 17

1. Aim

To design and implement a secure GitHub Pull Request workflow in which software changes are developed through feature branches and merged into the protected main branch only after peer review and approval.

2. Problem Statement

In collaborative software development, allowing developers to directly modify the main branch can introduce unreviewed or incorrect changes. The objective of this task is to create a secure branch-protection workflow using Git and GitHub, where developers create feature branches, commit their changes, raise Pull Requests, receive peer review and approval, and merge the approved changes into the protected main branch.

3. Problem Analysis

a) What is the engineering problem?

The engineering problem is to control how code changes are introduced into a shared Git repository. The workflow must prevent unauthorized or unreviewed direct changes to the main branch and ensure that changes pass through a Pull Request and peer-review process before merging.

b) Why is this problem complex?

☑ Requires multiple engineering concepts
☑ Requires analysis and decision making
☑ Involves design constraints
☑ Requires use of engineering tools
☑ Requires performance/security/safety consideration

4. Knowledge and Tools Required

Engineering concepts used:

1. Git version control
2. Git branching and merging
3. Pull Request and peer-review workflow
4. Branch protection
5. Collaborative software development

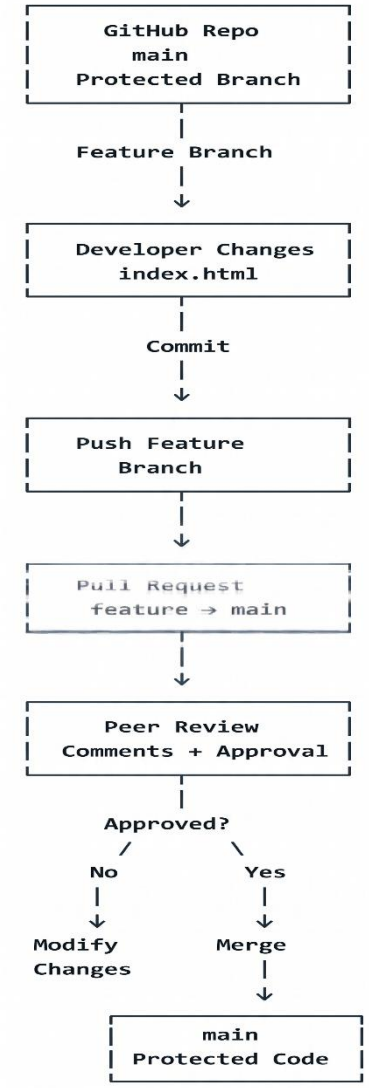
Tools/Software/Hardware used:

1. Git
2. GitHub
3. Visual Studio Code
4. Terminal

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5. Proposed Solution Design

Block Diagram / Architecture Diagram:

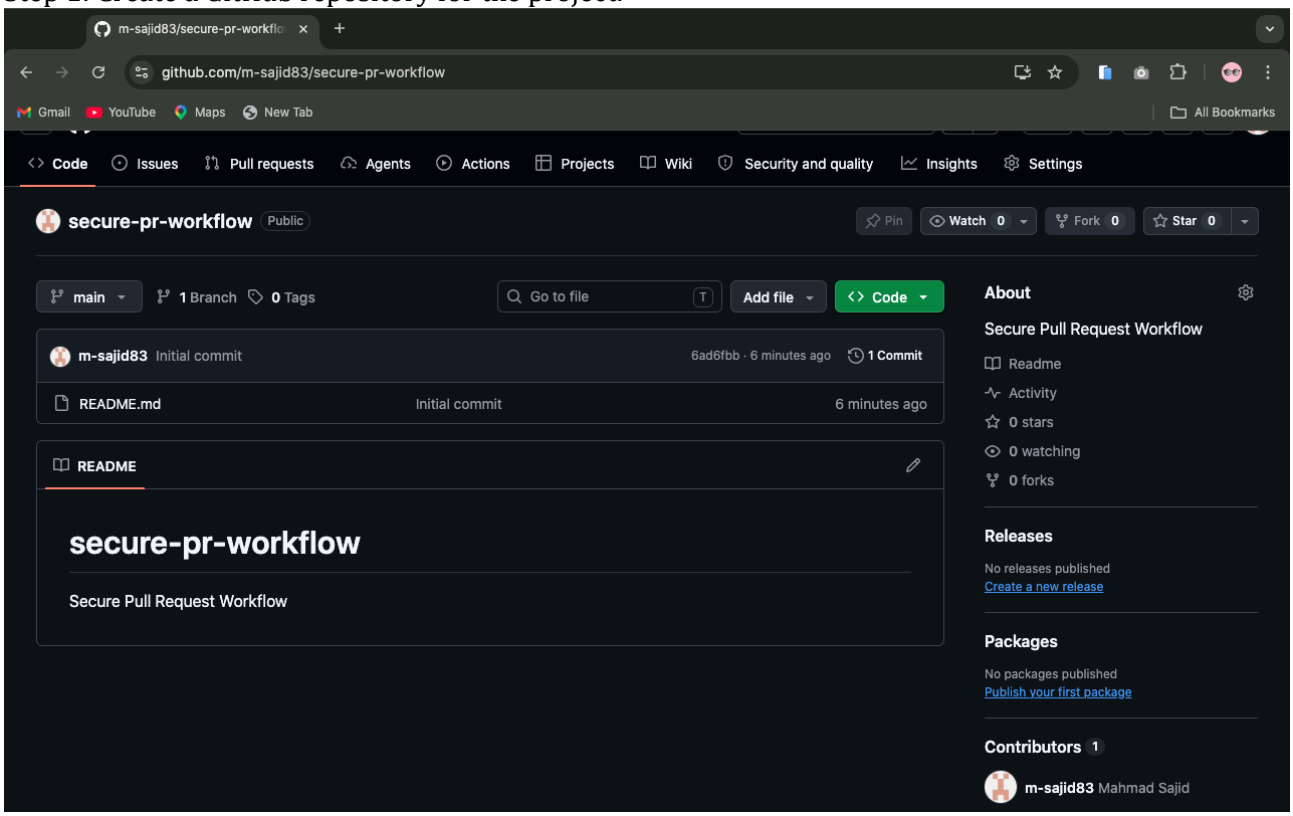


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Brief Working:
The proposed solution uses Git and GitHub to control changes to the main branch. The main branch is configured as a protected branch so that developers cannot directly introduce changes without following the required workflow. A developer creates a feature branch from main and makes the required changes. The changes are committed and pushed to GitHub. A Pull Request is then created from the feature branch to main. A peer reviews the changes and provides comments and approval. After approval, the Pull Request is merged into the protected main branch. This workflow improves code review, collaboration, traceability, and security.

6. Implementation / Procedure

Step 1: Create a GitHub repository for the project.



Step 2: Initialize/clone the repository and create the initial project files.

```
Last login: Mon Sep 21 20:50:32 on console
mac@macs-MacBook-Pro ~ % git clone https://github.com/m-sajid83/secure-pr-workflow.git
Cloning into 'secure-pr-workflow'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), done.
mac@macs-MacBook-Pro ~ % cd secure-pr-workflow
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch main
Your branch is up to date with 'origin/main'.

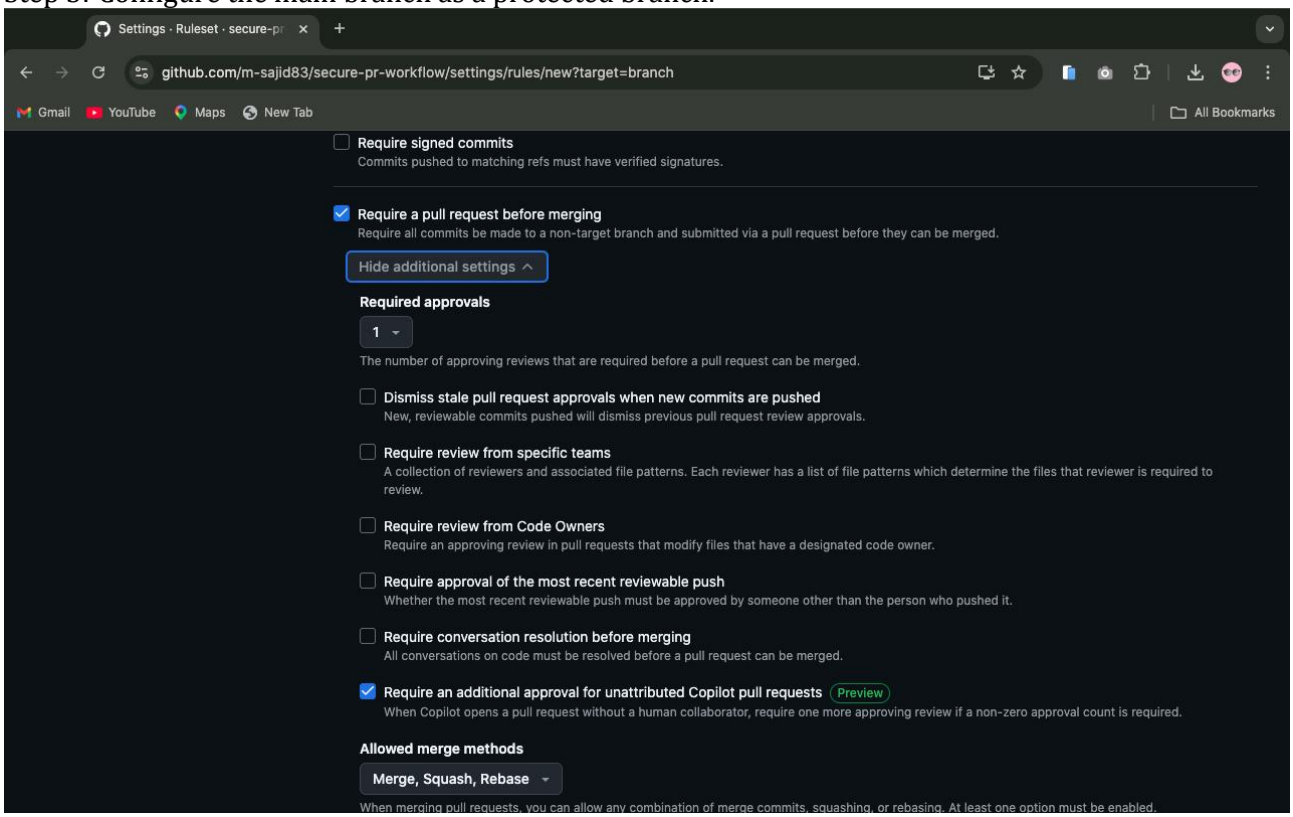
nothing to commit, working tree clean
mac@macs-MacBook-Pro secure-pr-workflow % code .
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
mac@macs-MacBook-Pro secure-pr-workflow % git add index.html
mac@macs-MacBook-Pro secure-pr-workflow % git commit -m "Add initial project"
[main b2000c8] Add initial project
 1 file changed, 10 insertions(+)
 create mode 100644 index.html
mac@macs-MacBook-Pro secure-pr-workflow % git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 436 bytes | 145.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
To https://github.com/m-sajid83/secure-pr-workflow.git
   6ad6fbb..b2000c8  main -> main
mac@macs-MacBook-Pro secure-pr-workflow %
```

COMPLEX ENGINEERING PROBLEM - REPORT

Step 3: Configure the main branch as a protected branch.



Step 4: Create the feature-homepage branch.

```
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git pull origin main
From https://github.com/m-sajid83/secure-pr-workflow
 * branch            main       -> FETCH_HEAD
Already up to date.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout -b feature-homepage
Switched to a new branch 'feature-homepage'
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
nothing to commit, working tree clean
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
mac@macs-MacBook-Pro secure-pr-workflow % git add index.html
mac@macs-MacBook-Pro secure-pr-workflow % git commit -m "Add project information"
[feature-homepage c4d41c5] Add project information
 1 file changed, 7 insertions(+), 1 deletion(-)
mac@macs-MacBook-Pro secure-pr-workflow % git log --oneline
c4d41c5 (HEAD -> feature-homepage) Add project information
b2000c8 (origin/main, origin/HEAD, main) Add initial project
6ad6fbb Initial commit
mac@macs-MacBook-Pro secure-pr-workflow % git push -u origin feature-homepage
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 502 bytes | 502.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'feature-homepage' on GitHub by visiting:
remote:      https://github.com/m-sajid83/secure-pr-workflow/pull/new/feature-homepage
remote:
To https://github.com/m-sajid83/secure-pr-workflow.git
 * [new branch]      feature-homepage -> feature-homepage
Branch 'feature-homepage' set up to track remote branch 'feature-homepage' from 'origin'.
mac@macs-MacBook-Pro secure-pr-workflow %
```

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Step 5: Modify the project file and commit the changes.

```
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git pull origin main
From https://github.com/m-sajid83/secure-pr-workflow
 * branch            main       -> FETCH_HEAD
Already up to date.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout -b feature-homepage
Switched to a new branch 'feature-homepage'
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
nothing to commit, working tree clean
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
mac@macs-MacBook-Pro secure-pr-workflow % git add index.html
mac@macs-MacBook-Pro secure-pr-workflow % git commit -m "Add project information"
[feature-homepage c4d41c5] Add project information
 1 file changed, 7 insertions(+), 1 deletion(-)
mac@macs-MacBook-Pro secure-pr-workflow % git log --oneline
c4d41c5 (HEAD -> feature-homepage) Add project information
b2000c8 (origin/main, origin/HEAD, main) Add initial project
6ad6fbb Initial commit
mac@macs-MacBook-Pro secure-pr-workflow % git push -u origin feature-homepage
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 502 bytes | 502.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'feature-homepage' on GitHub by visiting:
remote:      https://github.com/m-sajid83/secure-pr-workflow/pull/new/feature-homepage
remote:
To https://github.com/m-sajid83/secure-pr-workflow.git
 * [new branch]      feature-homepage -> feature-homepage
Branch 'feature-homepage' set up to track remote branch 'feature-homepage' from 'origin'.
mac@macs-MacBook-Pro secure-pr-workflow %
```

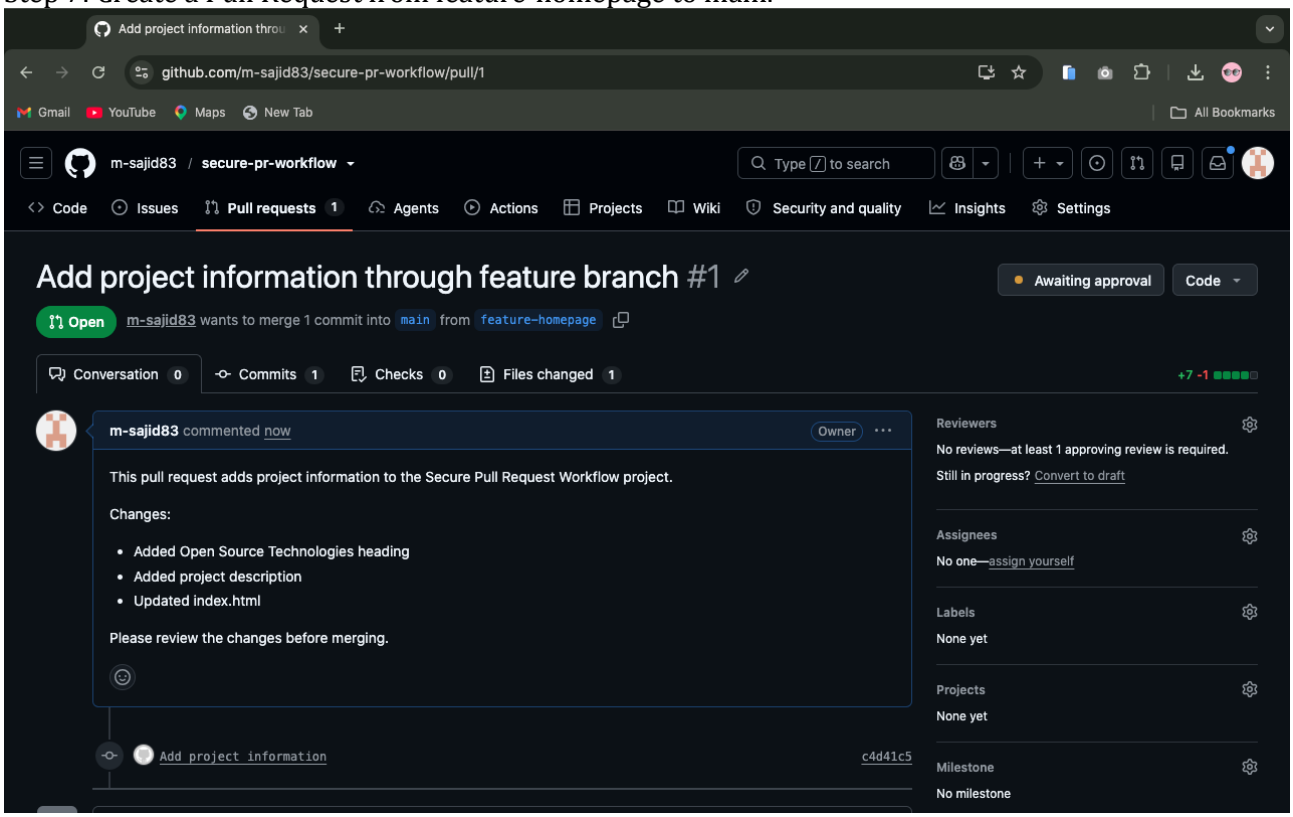
Step 6: Push the feature branch to GitHub.

```
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
mac@macs-MacBook-Pro secure-pr-workflow % git pull origin main
From https://github.com/m-sajid83/secure-pr-workflow
 * branch            main       -> FETCH_HEAD
Already up to date.
mac@macs-MacBook-Pro secure-pr-workflow % git checkout -b feature-homepage
Switched to a new branch 'feature-homepage'
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
nothing to commit, working tree clean
mac@macs-MacBook-Pro secure-pr-workflow % git status
On branch feature-homepage
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

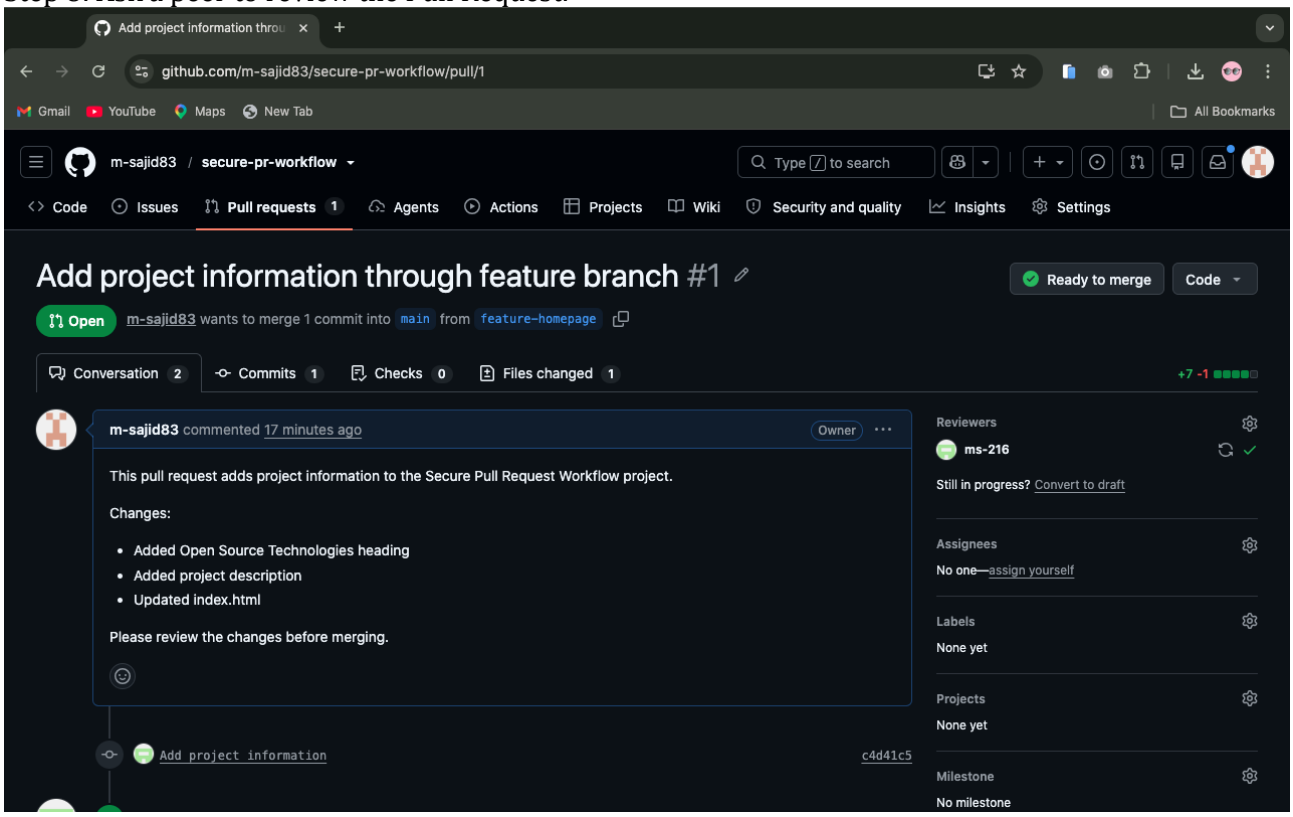
no changes added to commit (use "git add" and/or "git commit -a")
mac@macs-MacBook-Pro secure-pr-workflow % git add index.html
mac@macs-MacBook-Pro secure-pr-workflow % git commit -m "Add project information"
[feature-homepage c4d41c5] Add project information
 1 file changed, 7 insertions(+), 1 deletion(-)
mac@macs-MacBook-Pro secure-pr-workflow % git log --oneline
c4d41c5 (HEAD -> feature-homepage) Add project information
b2000c8 (origin/main, origin/HEAD, main) Add initial project
6ad6fbb Initial commit
mac@macs-MacBook-Pro secure-pr-workflow % git push -u origin feature-homepage
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
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Total 3 (delta 0), reused 0 (delta 0)
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remote:
To https://github.com/m-sajid83/secure-pr-workflow.git
 * [new branch]      feature-homepage -> feature-homepage
Branch 'feature-homepage' set up to track remote branch 'feature-homepage' from 'origin'.
mac@macs-MacBook-Pro secure-pr-workflow %
```

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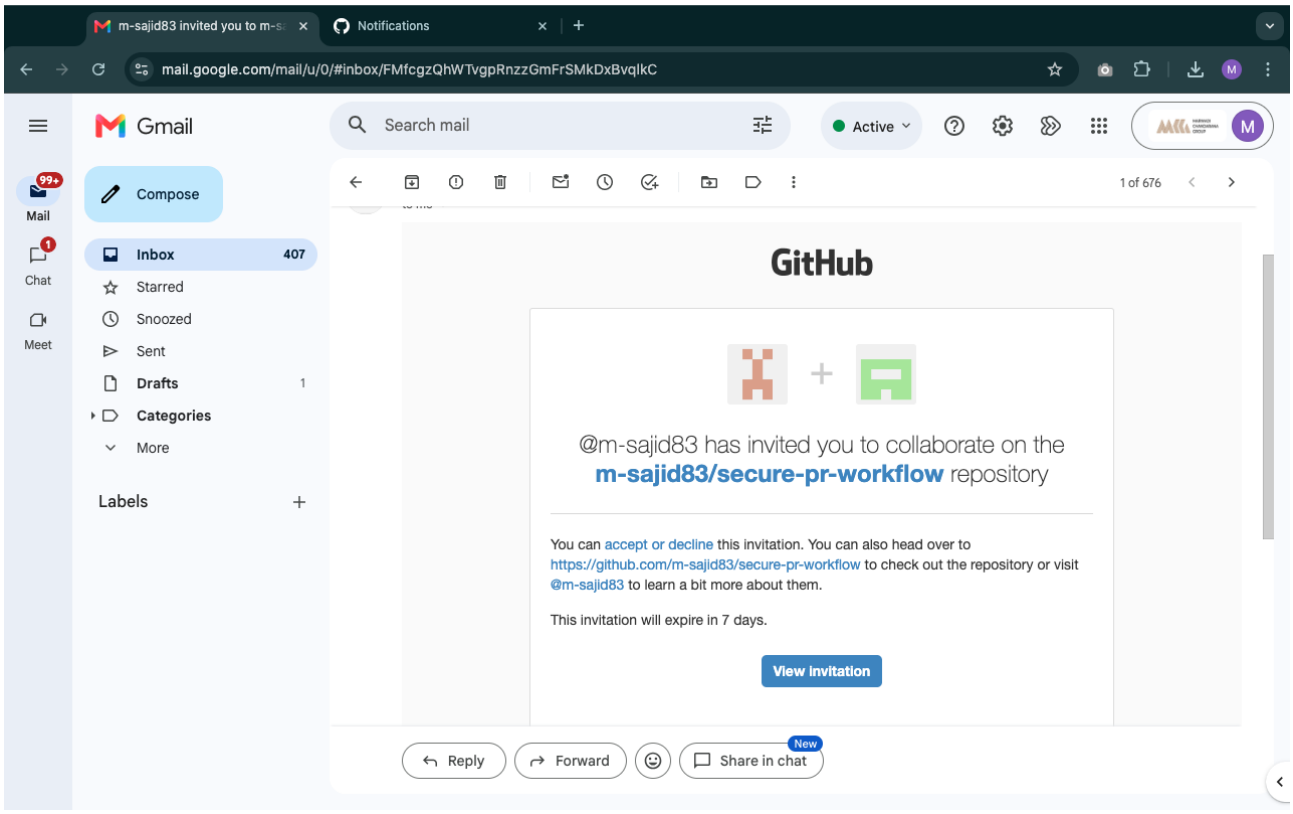
Step 7: Create a Pull Request from feature-homepage to main.



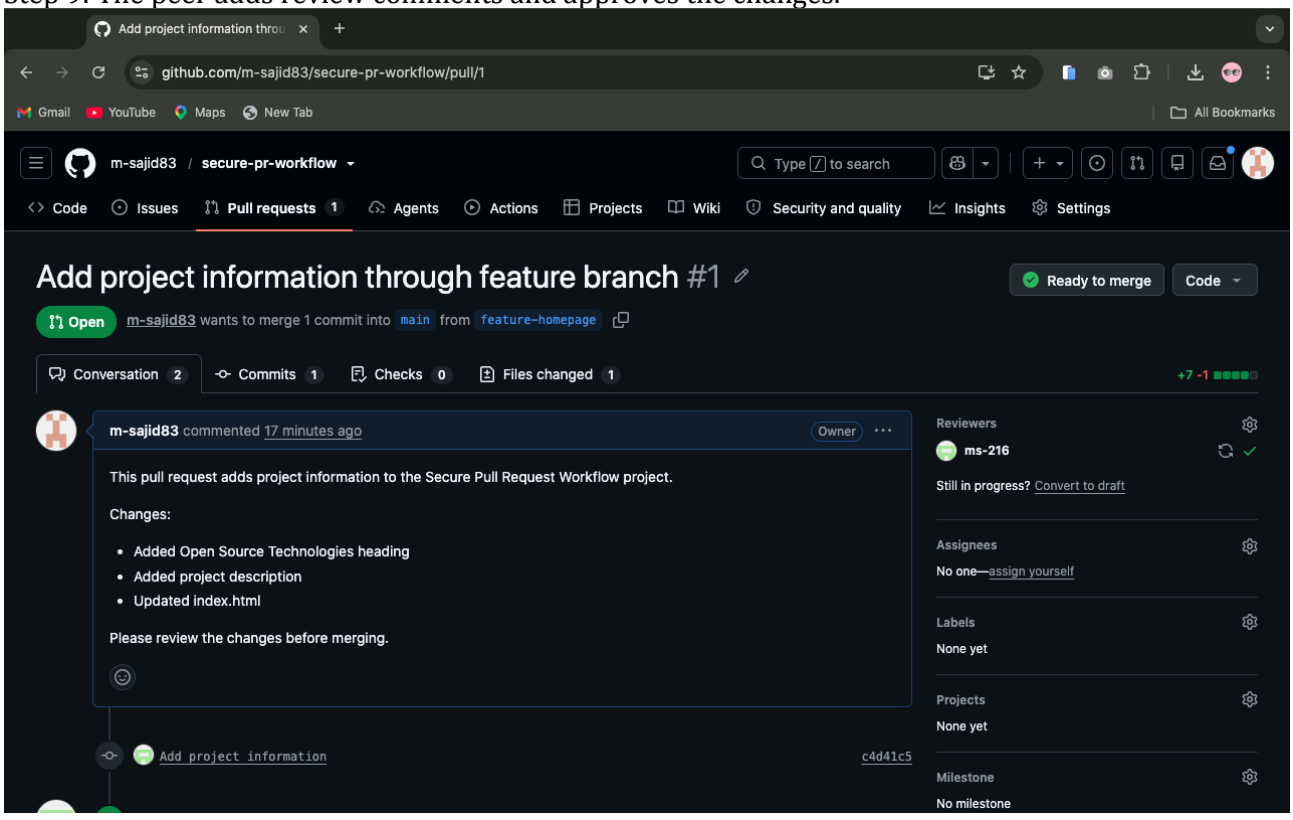
Step 8: Ask a peer to review the Pull Request.



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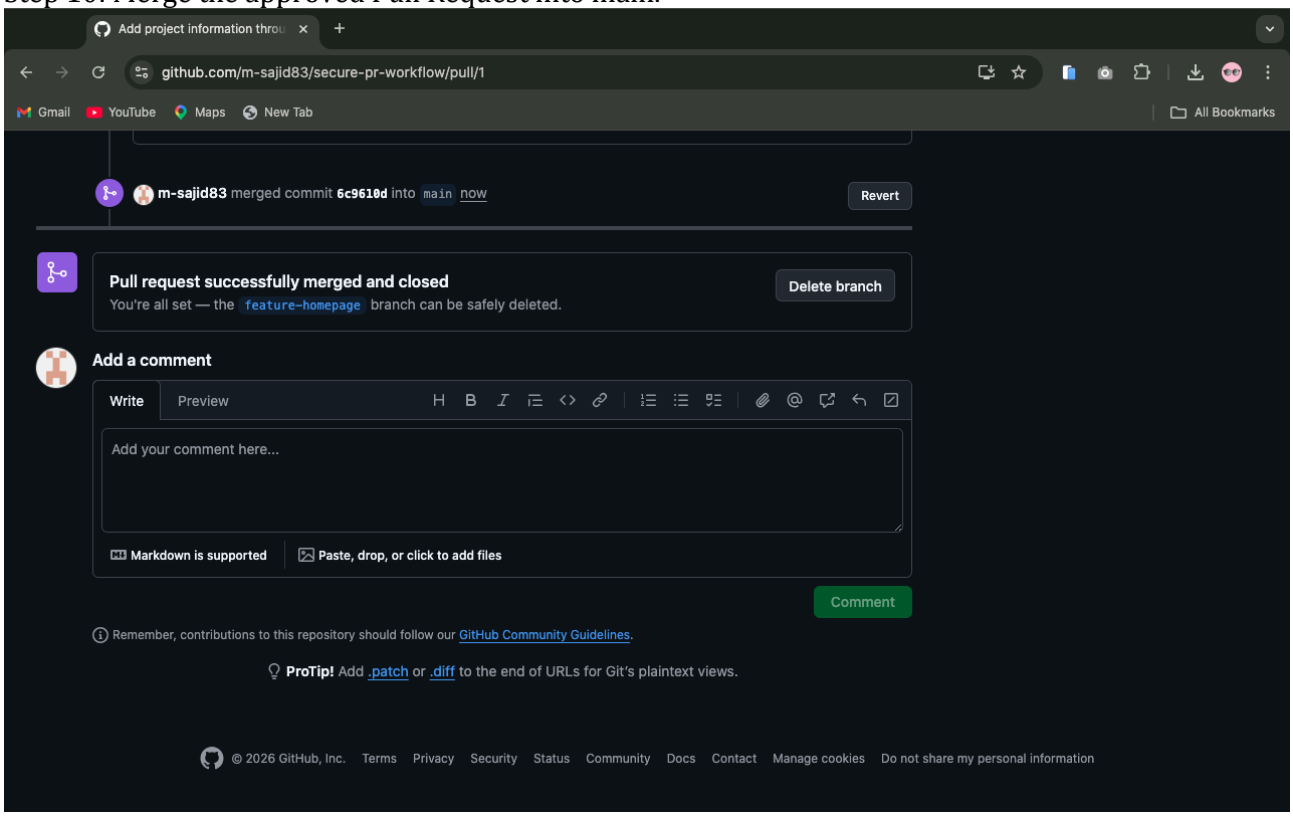


Step 9: The peer adds review comments and approves the changes.

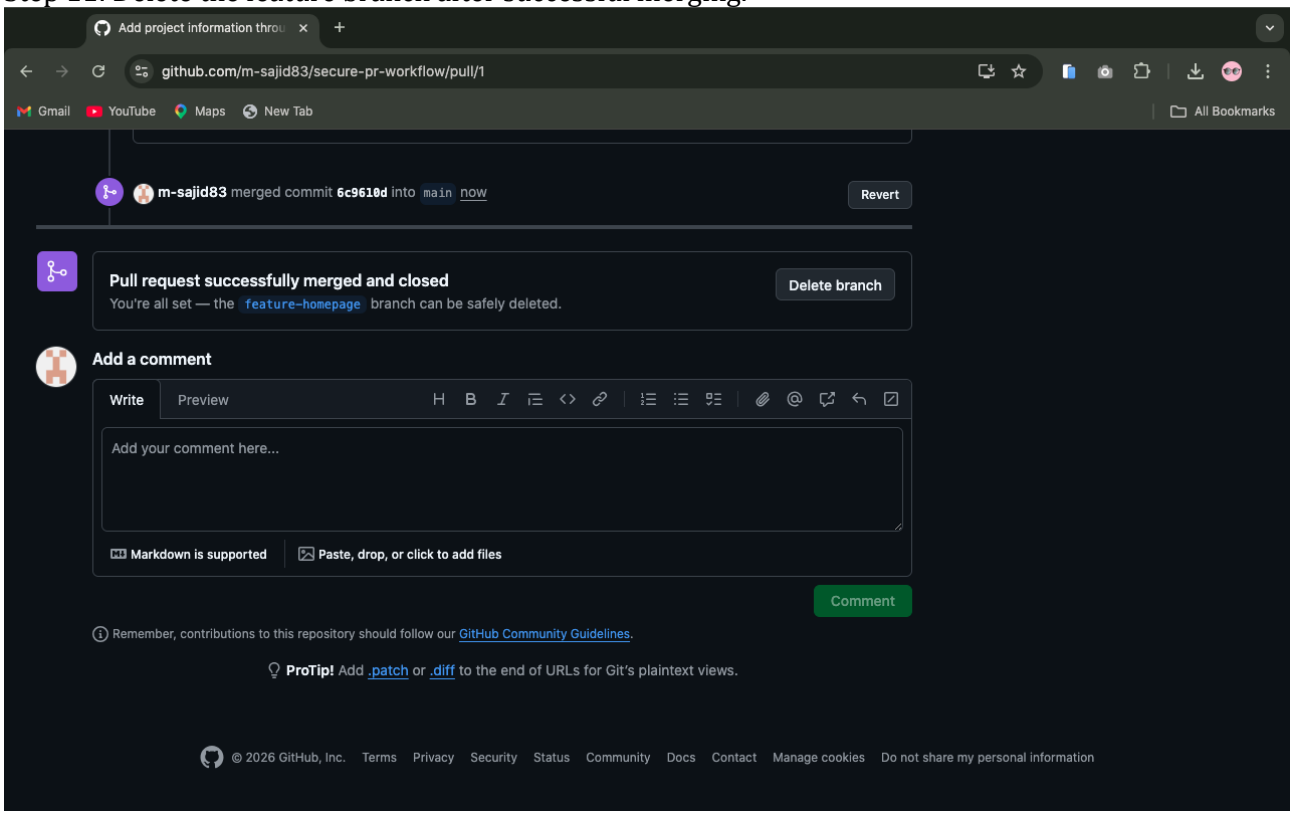


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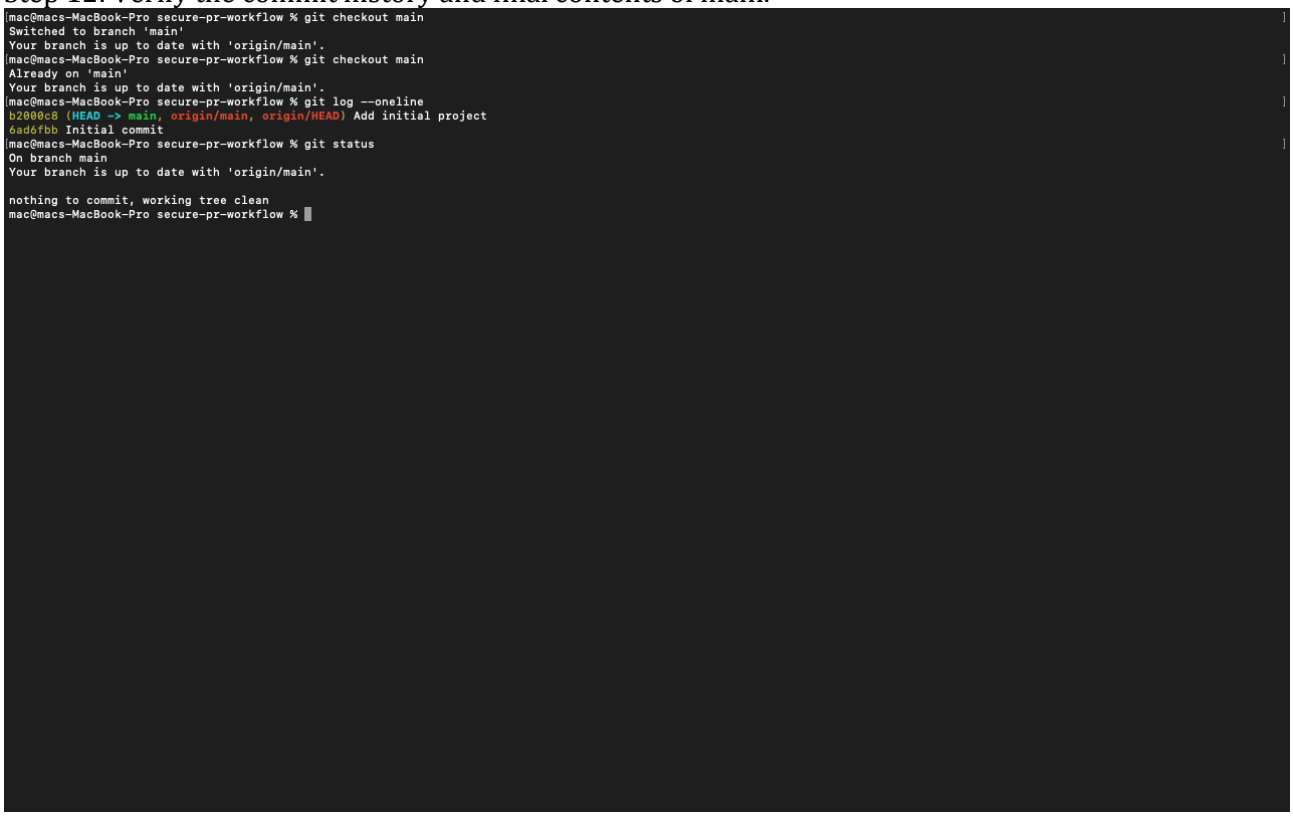
Step 10: Merge the approved Pull Request into main.



Step 11: Delete the feature branch after successful merging.



Step 12: Verify the commit history and final contents of main.



7. Testing and Results

Test/Parameter	Expected Result	Obtained Result
Feature-branch creation	Feature branch should be created	Feature branch created successfully
Feature commit	Changes should be committed	Changes committed successfully
Feature branch push	Branch should appear on GitHub	Branch appeared on GitHub
Pull Request	PR should be created from feature branch to main	Pull Request created successfully
Peer review	Reviewer should be able to comment and approve	Review completed and approved
Branch protection	Main should require PR/review	Main branch protected
Merge	Approved PR should merge into main	PR merged successfully
Final verification	Changes should appear in main	Changes successfully available in main

8. Challenges Faced and Solutions

Challenge	Solution
Understanding feature branch workflow	Practiced branch creation, commit and push commands
Preventing direct changes to main	Configured branch protection
Understanding Pull Request process	Created a PR from feature branch to main
Peer review coordination	Used a second group member as reviewer
Verifying successful merge	Checked Pull Request status and commit history

9. Conclusion

The Secure Pull Request Workflow was successfully implemented using Git and GitHub. The project demonstrated the use of feature branches, commits, Pull Requests, peer review, approval and controlled merging. The main branch was protected so that changes followed the defined review workflow instead of being directly introduced. The process improved collaboration, traceability and control over changes. The final Pull Request was reviewed, approved and successfully merged into the main branch.

10. SDG Mapping and Justification

SDG 16: Peace, Justice and Strong Institutions
Promotes controlled, accountable and traceable software changes through branch protection, Pull Requests, peer review and approval.

SDG 17: Partnerships for the Goals
Supports collaboration between developers through feature branches, Pull Requests and peer review.

11. References

[1] GitHub, “About pull requests,” GitHub Documentation.

[2] GitHub, “Managing a branch protection rule,” GitHub Documentation.

[3] Git, “Git Documentation,” Git SCM Documentation.